ApeX-Pro ECDL Laser Series

Product Overview | 780-795 nm External Cavity Diode Laser

Product Description

The ApeX-Pro is a compact External Cavity Diode Laser (ECDL) operating at 780 nm and 795 targeting both the Rubidium D1 and D2 transitions. This ECDL is based on a Littrow configuration with a piezo-driven diffraction grating, it provides single-mode output with a Lorentzian linewidth better than 1 MHz and mode-hop-free tuning more than 20 GHz.
Three variants cover the most common laboratory requirements: a basic free-space output, a free-space model with an integrated optical isolator, and a fibre-coupled model with PM fibre output. All models connect to the ApeX-Ctrl Laser Controller (QPEC-010).

Key Features

- Linewidth < 1 MHz
- Mode-hop-free tuning (PZT + Current) > 20 GHz
- Mode-hop-free tuning (PZT) >3 GHz
- Coarse tuning range 780– 795nm
- Output power > 80 mW (free space)
- Optional integrated isolator > 55 dB
- Optional PM fibre output (FC/APC)

Models

Model No.	Description
780AP15-FS	ApeX-Pro, Free Space
780AP15-FS-IS	ApeX-Pro, Free Space + Isolator
780AP15-FC-IS	ApeX-Pro, Fibre Coupled + Isolator
QPEC-010	ApeX-Ctrl Laser Controller[1]

Typical Applications

- Laser cooling & trapping (MOT)
- Bose-Einstein condensates
- Atom interferometry
- Saturated absorption spectroscopy
- Optical frequency reference (when locked)
- Quantum information / QKD
- Rydberg atom experiments
- High-res atomic/molecular spectroscopy



ApeX-Pro

[1] *Comes together with all models of ApeX-Pro; See details below.*

Models

Parameter	All Models
Typical Wavelength (nm)	780
Coarse Tuning Range (nm)	780–795
Output Power (mW)	> 80
Mode-Hop-Free Tuning (GHz)	> 20
Linewidth (MHz)	< 1
Optical Isolation (dB)	>55
Output Format	Free Space
Polarization Extinction Ratio	> 100:1
Beam profile (mm)	2:3
Operating Temperature (°C)	25 – 80
Dimensions L×W×H (mm)	175×90×55
Weight (g)	~ 600
Interface Connector	DSUB-15

ApeX-Ctrl Laser Controller

Model QPEC-010 – Precision Current, Temperature & Piezo Controller

Product Description

The ApeX-Ctrl (QPEC-010) is a compact bench-top laser controller that integrates precision current driving, TEC-based temperature control, and high-voltage piezo actuation within a single 3U half-rack unit. The front panel features two BNC I/O connectors for external current and piezo scan/modulation/control. The rear panel includes a USB interface for communication and a DB15 connector for direct integration with ApeX-Pro lasers. This unit is compatible with all ApeX-Pro laser models.



ApeX-Ctrl

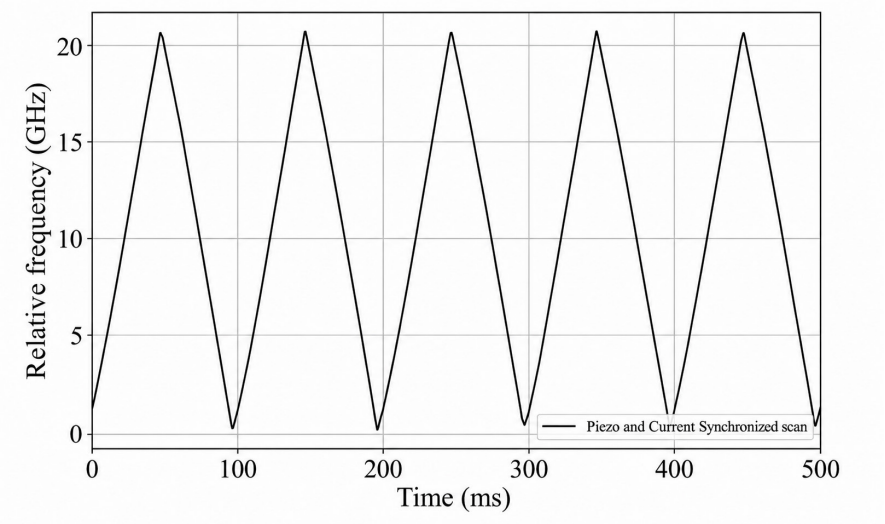
Models

Parameter	Value
Current Controller	
Output Range	0 – 200 mA
Current Soft Limit	Programmable, 10–250 mA
Temperature Controller	
Control Range	25 – 80°C
Stability (±)	0.005 °C
Piezo Driver	
Output Voltage	0 – 50 V
Scan Waveforms (external)	Ramp, triangle, sine
Interface & General	
Front Panel I/O	2× BNC: Scan In (Piezo and current)
Rear Connectivity	USB-B, RS-232 (DB9), DC Power
Software	ApexPro
Dimensions L×W×H (mm)	320 × 230 × 110
Weight	~ 2.5 kg

Performance report:

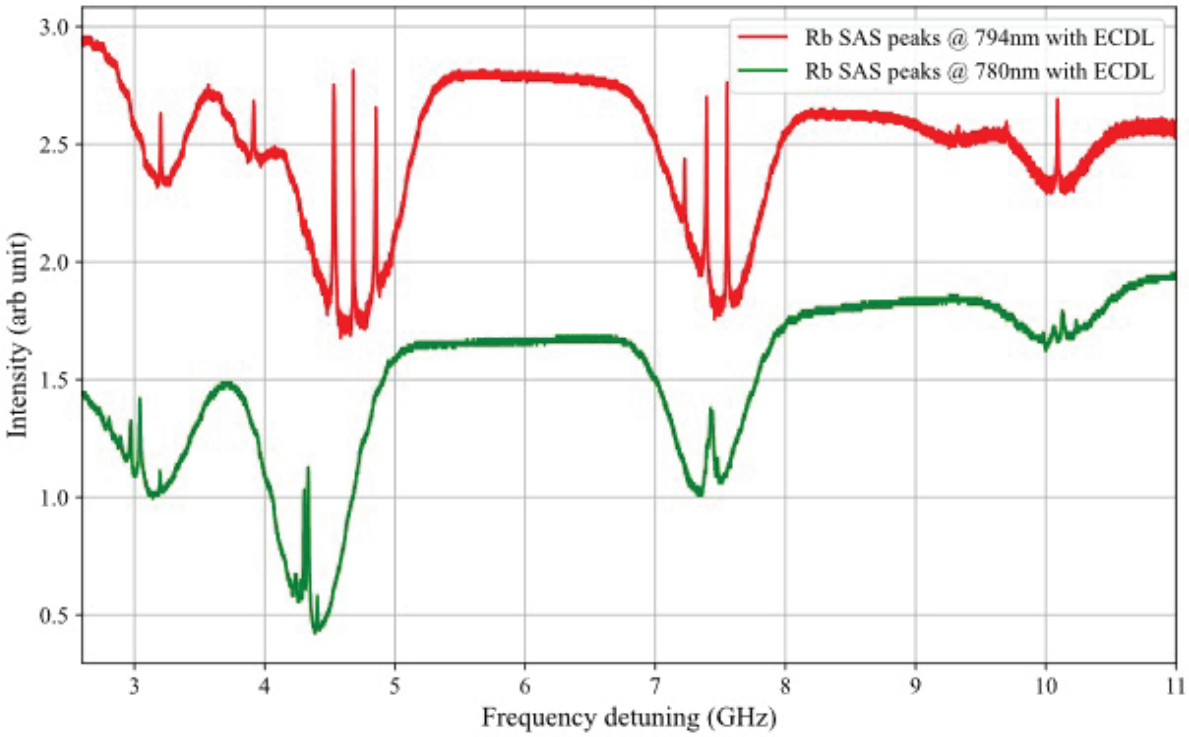
Mode hop free scan:

We are able to achieve a mode-hop-free scan of more than 20 GHz with our ECDL with a synchronous scan using current and the piezo (grating). The following plot shows a triangular scan recorded using the wavemeter (HighFinesse WS-6 with resolution better than 4 MHz). The y-axis represents the relative frequency of the scan, and the x-axis represents time.



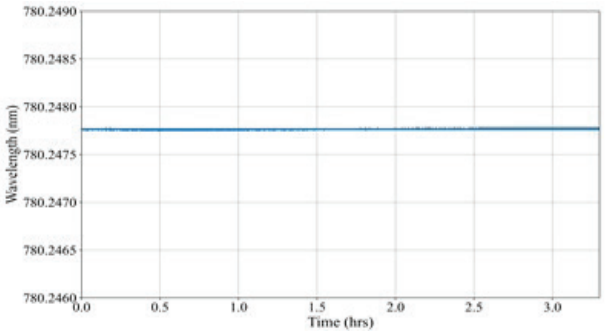
Saturated absorption spectroscopy (SAS) measurement for Rb D1 and D2 lines:

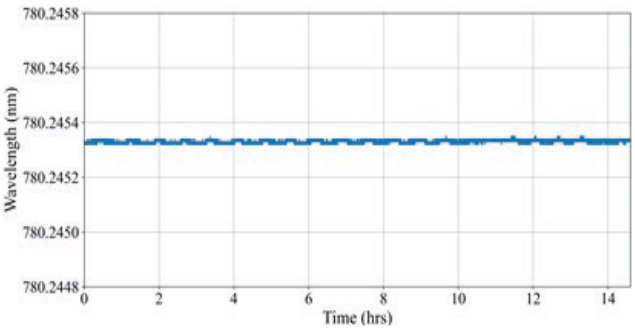
As shown in the figure below, our ECDL clearly resolves the hyperfine saturated absorption spectroscopy (SAS) spectra of both Rb-85 and Rb-87 isotopes for the D1 (794 nm) and D2 (780 nm) transitions. The red trace corresponds to the D1 SAS spectrum, whereas the green trace corresponds to the D2 SAS spectrum of Rb.



Locking performance:

We tested our ECDL by locking it to one of the hyperfine peaks of the Rb D2 line with our control electronics box. Repeated multiple times across different peaks, following two adjacent plots demonstrate the robustness of the ECDL for short (3 hrs) and long (15 hrs) durations . The data was recorded on wavemeter (HighFinesse WS-6 200 with frequency resolution <4 MHz)





www.thequantum.ai

Suite 305B, Capital Park, Image Garden Rd, HITEC City, Hyderabad, India,Call : +91 81210 27006

Please Contact Info@thequantum.ai for further Information